

An Intrinsic-Dimensional Jackson Bound for Prefix-Tuning a Fixed Attention Head

Candidate full solution packet for arXiv:2402.14753

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Status. Candidate full affirmative answer; likely valid, pending human review. For geodesically Lipschitz maps on $\mathbb{S}^m$, one finite fixed attention head has a prefix of length $O_m((L/\epsilon)^m)$ giving uniform error ϵ . Thus the exponent agrees with the trained-network benchmark named in the source, rather than the source construction’s exponent $10 + 14m + 4m^2$.

1 Source question

Petrov, Torr, and Bibi prove in *Prompting a Pretrained Transformer Can Be a Universal Approximator*, arXiv:2402.14753 [1], that a specially constructed fixed attention head is universal under prefix-tuning. For Lipschitz maps on $\mathbb{S}^m$ their construction uses

$$N = O\left(\epsilon^{-10-14m-4m^2}\right)$$

prefix positions, whereas they quote $O(\epsilon^{-m})$ trainable parameters as the usual benchmark for trained networks. They ask whether a tighter Jackson bound exists, or whether prefix-tuning and prompting inherently lose asymptotic efficiency.

We prove that the tighter bound exists in the same idealized fixed-head model. The proof is direct: use a geodesic net as the keys, sampled target values as the values, and make softmax localize near a closest net point. This avoids both the source’s spherical convolution and its uniform Riemann-sum approximation.

2 Model and statement

Let $\rho(x, y) = \arccos \langle x, y \rangle$ be geodesic distance on the unit sphere $\mathbb{S}^m \subset \mathbb{R}^{m+1}$. Write A_j for the surface area of $\mathbb{S}^j$ and set

$$C_m = \frac{2m\pi^{m-1}A_m}{A_{m-1}}. \tag{1}$$

For a sequence $X = (x_1, \dots, x_T)$ and prefix $P = (p_1, \dots, p_N)$, the source’s classical attention head is

$$[u(X; P)]_k = \frac{\sum_{i=1}^N e^{x_k^\top H p_i} W_V p_i + \sum_{j=1}^T e^{x_k^\top H x_j} W_V x_j}{\sum_{i=1}^N e^{x_k^\top H p_i} + \sum_{j=1}^T e^{x_k^\top H x_j}}. \tag{2}$$

Theorem 1 (Benchmark-rate fixed-head Jackson bound). *Fix $m \geq 1$ and $T \geq 1$. There are matrices $H, W_V \in \mathbb{R}^{3(m+1) \times 3(m+1)}$, independent of the target, accuracy, prefix length, and input sequence, with the following property.*

Prefix-Tuning and Prompting a Pretrained Transformer might be Less efficient than Training it Typically, with neural networks one expects that the number of trainable parameters would grow as $\mathcal{O}(\epsilon^{-m})$ (Schmidt-Hieber, 2021). Indeed that is the case for universal approximation with a transformer when one learns the key, query and value matrices and the MLP parameters as shown by Yun et al. (2019). However, as Equation (7) shows, our construction results in the trainable parameters (prefix length in our case) growing as $\mathcal{O}(\epsilon^{-10-14m-4m^2})$. That the m^2 term indicates worse asymptotic efficiency of prefix-tuning and prompting compared to training a transformer. However, our approach may not be tight. Thus, it remains an open question if a tighter Jackson bound exists or if prefix-tuning and prompting inherently require more trainable parameters to reach the same approximation accuracy as training a transformer.

Figure 1: The exact open question on source PDF page 9.

Let $f : \mathbb{S}^m \rightarrow \mathbb{R}^{m+1}$ satisfy

$$\|f(x) - f(y)\|_2 \leq L\rho(x, y), \quad x, y \in \mathbb{S}^m,$$

where $L > 0$, and put $B = \|f\|_\infty$. For every $\epsilon > 0$ define

$$\delta = \min \left\{ \frac{1}{2}, \frac{\epsilon}{6L} \right\}.$$

There is a prefix of length

$$N \leq C_m \delta^{-m} \tag{3}$$

such that, after the fixed embedding $\Pi x = (x, 0, 0) \in \mathbb{R}^{3(m+1)}$ and projection onto the first block,

$$\sup_{(x_1, \dots, x_T) \in (\mathbb{S}^m)^T} \max_{1 \leq k \leq T} \|\Pi^{-1}[u(\Pi X; P)]_k - f(x_k)\|_2 \leq \epsilon. \tag{4}$$

Consequently, for fixed m, L, T and f ,

$$N = O_m(\epsilon^{-m}) \quad (\epsilon \downarrow 0).$$

Each prefix position has $3(m+1)$ coordinates, so the number of trainable prefix coordinates has the same ϵ -exponent m .

If $L = 0$, the map is constant. It is represented exactly with one prefix: take zero key, value $(T+1)f$, and note that all $T+1$ attention scores are zero. Thus this harmless case is omitted from the quantitative statement.

3 A small intrinsic net

Lemma 2. For $0 < \delta \leq 1/2$, the sphere has a geodesic δ -net $\{b_1, \dots, b_N\}$ satisfying (3).

Proof. Choose a maximal geodesic δ -separated set. Maximality makes it a δ -net, while its open caps of radius $\delta/2$ are disjoint. A cap of radius $\delta/2$ has area

$$A_{m-1} \int_0^{\delta/2} \sin^{m-1} t \, dt.$$

Since $\sin t \geq 2t/\pi$ on $[0, \pi/2]$, this is at least

$$A_{m-1} \left(\frac{2}{\pi} \right)^{m-1} \frac{(\delta/2)^m}{m} = \frac{A_{m-1} \delta^m}{2m\pi^{m-1}}.$$

The disjoint caps fit inside a sphere of area A_m . Rearrangement gives $N \leq C_m \delta^{-m}$. $\square$

The exponent is m , the intrinsic dimension of the domain, rather than the ambient dimension $m+1$.

4 Softmax is a local interpolant

Fix the net in Lemma 2. For $\lambda > 0$ define

$$g(x) = \frac{\sum_{i=1}^N e^{\lambda \langle x, b_i \rangle} f(b_i)}{\sum_{i=1}^N e^{\lambda \langle x, b_i \rangle}}. \quad (5)$$

Lemma 3. If

$$\lambda \geq 2\delta^{-2} \log \left(\frac{\pi N}{\delta} \right), \quad (6)$$

then $\sup_{x \in \mathbb{S}^m} \|g(x) - f(x)\|_2 \leq 3L\delta$.

Proof. Fix x and choose b_j with $\rho(x, b_j) \leq \delta$. A net point with $\rho(x, b_i) \geq 2\delta$ has score at most $\lambda \cos(2\delta)$, whereas the denominator in (5) is at least $e^{\lambda \cos \delta}$. Moreover,

$$\begin{aligned} \cos \delta - \cos(2\delta) &= 2 \sin(3\delta/2) \sin(\delta/2) \\ &\geq \frac{6}{\pi^2} \delta^2 \geq \frac{1}{2} \delta^2, \end{aligned}$$

where both sine arguments lie in $[0, \pi/2]$. Hence the total softmax mass q_x assigned outside the 2δ cap obeys

$$q_x \leq N e^{-\lambda \delta^2/2} \leq \frac{\delta}{\pi}.$$

Inside the cap, Lipschitzness gives error at most $2L\delta$; everywhere on the sphere it gives error at most πL . Since (5) is a convex combination,

$$\|g(x) - f(x)\|_2 \leq 2L\delta + \pi L q_x \leq 3L\delta.$$

The estimate is uniform in x . $\square$

5 Realization by one finite fixed head

Use three blocks of size $m + 1$ and fix

$$H = \begin{bmatrix} 0 & I & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad W_V = \begin{bmatrix} 0 & 0 & I \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (7)$$

For each net point put

$$p_i = \begin{bmatrix} 0 \\ \lambda b_i \\ f(b_i) \end{bmatrix}. \quad (8)$$

Then, for every x_k , the projected prefix score is $\lambda \langle x_k, b_i \rangle$ and the projected prefix value is $f(b_i)$. Every embedded input token has score zero and projected value zero. Therefore

$$\Pi^{-1}[u(\Pi X; P)]_k = \frac{\sum_i e^{\lambda \langle x_k, b_i \rangle} f(b_i)}{\sum_i e^{\lambda \langle x_k, b_i \rangle} + T}. \quad (9)$$

This identity uses ordinary finite matrices; no limit is hidden in it.

Let $D(x) = \sum_i e^{\lambda \langle x, b_i \rangle}$. Since some b_j is within δ of x , $D(x) \geq e^{\lambda \cos \delta}$. Equation (5) and the convexity bound on its numerator give

$$\|\Pi^{-1}u(\Pi X; P)_k - g(x_k)\|_2 \leq \frac{BT}{D(x_k)} \leq BT e^{-\lambda \cos \delta}. \quad (10)$$

Choose λ satisfying (6) and, in addition,

$$\lambda \geq \frac{1}{\cos \delta} \max \left\{ 0, \log \left(\frac{2BT}{\epsilon} \right) \right\}, \quad (11)$$

with the second term read as zero when $B = 0$. Lemma 3, the definition of δ , and (10) now give

$$\|\Pi^{-1}u(\Pi X; P)_k - f(x_k)\|_2 \leq 3L\delta + \frac{\epsilon}{2} \leq \epsilon.$$

This proves Theorem 1.

Arbitrary continuous targets. The same proof also removes the source’s harmonic-coefficient hypothesis and restriction $m \geq 8$. If f is merely continuous, let $\omega_f(r) = \sup_{\rho(x,y) \leq r} \|f(x) - f(y)\|_2$. The near contribution in Lemma 3 is $\omega_f(2\delta)$ and the far contribution is at most $2Bq_x$. First choose δ from the modulus and then λ ; this gives density for every $m \geq 1$ with the same finite head. Lipschitzness is needed only to turn this into the explicit Jackson rate (3).

6 What the open question receives

	Source construction [1]	This packet
Prefix exponent	$10 + 14m + 4m^2$	m
Sphere dimension	$m \geq 8$	$m \geq 1$
Regularity artefact	extra harmonic constant	none
Pretrained head	limiting block constant	finite fixed matrices
Sequence length T	element-wise extension	same N ; T enters λ

For fixed m , the number of trainable prefix coordinates is a constant multiple of N , hence $O(\epsilon^{-m})$. This answers the affirmative branch of the source question: the much larger exponent is an artefact of the particular proof, not an inherent prefix-length penalty in its constructed fixed-head model. Via the source’s reduction from prefix-tuning to soft prompting, the same conclusion applies to that idealized soft-prompt setting.

There is also a small repair to the source’s conversion. Its proof of exact split-to-classical realization takes a block constant M “tending to $-\infty$ ”. A finite M does not erase the input denominator exactly. Equations (7)–(11) instead retain those terms, calculate them exactly, and absorb them without increasing N .

7 Novelty check and limitations

A bounded search through 11 August 2026 found two close later results. Hu et al. [2] prove prompt-tuning universality for a one-head, one-attention-layer transformer, but use quantization, additional feed-forward layers, and a long prompt indexing a discretized function class; they do not give this uniform intrinsic-dimensional prefix Jackson bound. Liu et al. [3, Theorem A.1] give an $O(dnN_X)$ covering-number parameter bound, but their target-dependent parameters occur in a trainable linear layer and trainable attention matrices, not solely in prefixes of a frozen head. Exact-phrase, title-citation, and architecture-keyword searches found no later proof of the statement here.

The theorem concerns the specially designed frozen head in [1], not an arbitrary transformer obtained from real-world pretraining. Prefix magnitudes are not bounded: (6) permits $\lambda = O(\epsilon^{-2} \log(1/\epsilon))$. The result counts real trainable coordinates, as the source question does, and does not measure their bit complexity or optimization difficulty. It also supplies no minimax lower bound for the complete unbounded-prefix hypothesis class. Thus the rate is called *benchmark-rate*, not formally minimax-optimal.

References

- [1] A. Petrov, P. H. S. Torr, and A. Bibi, *Prompting a Pretrained Transformer Can Be a Universal Approximator*, ICML 2024; arXiv:2402.14753.
- [2] J. Y.-C. Hu, W.-P. Wang, A. Gilani, C. Li, Z. Song, and H. Liu, *Fundamental Limits of Prompt Tuning Transformers: Universality, Capacity and Efficiency*, ICLR 2025; arXiv:2411.16525.
- [3] H. Liu, J. Y.-C. Hu, Z. Song, and H. Liu, *Attention Mechanism, Max-Affine Partition, and Universal Approximation*, NeurIPS 2025; arXiv:2504.19901.